

and G.655 fibres.

WDM MUX and DMUX. Wavelength division multiplexer/de-multiplexer combines/separates discrete signals transmitted at different wavelengths. Essentially, the multiplexer is a mixer and the de-multiplexer is a wavelength filter.

Wavelength-division de-multiplexers use several methods (bulk optical diffraction gratings or interference filters for spacing between 1nm and 2nm or fibre Bragg gratings (FBGs) for spacing less than 0.8nm) to separate different wavelengths depending on the spacing between the wavelengths. FBGs are of critical importance in DWDM systems in which multiple closely-spaced wavelengths require separation.

Light entering the FBG is diffracted by the induced period variations in the index of refraction. By spacing the periodic variations at multiples of the half-wavelength of the desired signal, each variation reflects light with a 360-degree phase shift, causing a constructive interference of a very specific wavelength while allowing others to pass.

Optical amplifier. In DWDM system, a pre-amplifier is used to boost the signal pulses at the receiver's side and a post-amplifier boosts signal at the transmitter's side. In-line amplifiers are placed at different points (usually every 50-60km) from the source to provide recovery of the signal before it is degraded significantly. EDFA is the most popular optical amplifier used to boost the signal level in the 1530nm to 1570nm region of the spectrum.

Most importantly, EDFAs allow signals to be regenerated without having to be converted back to electrical signals. In this amplifier, an external laser source of either 980nm or 1480nm is used to pump it and considerably high amplification of the order of 30dB is achieved.

Regenerator. If the length to be bridged is too long to be covered, regenerators are employed. These regenerators perform real 3R functions—reshaping, retiming and amplification of the signal. Therefore the signals have to be de-multiplexed, electrically regenerated and multiplexed again.

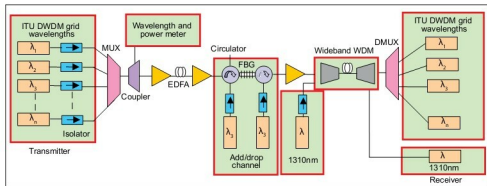


Fig. 3. A typical DWDM system

Optical add/drop multiplexer. In a DWDM system, optical add/drop multiplexer is used to add or drop some specific wavelengths from the already multiplexed optical signal. Such add/drop multiplexer may use complete de-multiplexing or other techniques. A fibre-optic circulator employing FBG may be used to drop any specific wavelength and pass the others.

DWDM system

DWDM is an optical multiplexing technology that allows many discrete transport channels, by combining and transmitting multiple signals simultaneously at different wavelengths, on the same fibre. In effect, one fibre is transformed into multiple virtual fibres. So, if you were to multiplex 32 STM-16 signals into one fibre, you would increase the carrying capacity of that fibre from 2.5Gbps to 80Gbps.

Because of DWDM, single fibres have been able to transmit data at speeds up to 400Gbps. A key advantage to DWDM is that it is protocol and bit-rate independent. DWDM-based networks can carry different types of traffic at different speeds over an optical channel, and so can transmit data in SDH, IP, ATM and Ethernet.

Such a DWDM system is shown in Fig. 3 where DWDM and EDFA is used to transmit several channels of high-bandwidth information over a single fibre. These signals are converted into corresponding wavelengths of light in accordance with the ITU grid and are multiplexed with the help of multichannel coupler or wavelength division multiplexer.

In order to minimise back reflec-

tions, an optical isolator is used with each optical source. Multiplexed optical signal is then routed through a single optical fibre cable. After covering a distance of 50-100km, the signal needs to be amplified. So, several stages of amplification using EDFA are required in order to cover long distances.

In order to add or drop some channels, an optical circulator in conjunction with FBG is used. After amplification, a wideband wavelength division multiplexer is used to combine 1310nm signal with the 1550nm window signals. At the receiver's side, 1310nm signal is extracted via another wideband wavelength division de-multiplexer.

Finally, a dense wavelength division de-multiplexer that is made of an array of FBGs, each tuned to specific transmission wavelength, separates all the 1550nm window signals. A tap coupler is used to couple some amount of optical power for wavelength and power measurement.

Conclusion

The growth of the fibre-optics industry, including the advances in optical sources and detectors that operate in the 1550nm range, the developments of EDFAs, FBG and multi-channel DWDM systems, has been explosive over the past five years. Presently, DWDM is one of the hottest emerging market segments in global telecommunications to usher in the next generation of services. ●

The author, currently with Bharat Sanchar Nigam Limited, is a recipient of senior research fellowship of UGC. His research interests include wired and wireless technologies for high-speed Internet access